Einstein also knew that the idea of photons, taken together with locality, implied a serious revision of Maxwell's laws of electromagnetism, as early as 1909. See Bacciagaluppi and Valentini 2009.

"contradiction with the principle [i.e., the special theory] of relativity": Bacciagaluppi and Valentini 2009, p. 487.

"Mr. de Broglie is right": Ibid., p. 487.

"No doubt it is my fault": Ibid., pp. 487–488. Bacciagaluppi and Valentini make this point themselves: "Einstein's argument is so concise that its point is easily missed, and one might well dismiss it as arising from an elementary confusion about the nature of probability" (p. 195).

confused about the nature of probability: After all, if the wave function is simply a statement about the probability that a single electron will be registered at a location on the film, then it's logically impossible for one electron's wave function to lead the film to register two electrons in two different locations. But that argument is circular reasoning—literal question begging—because it already assumes that the wave function is merely a probability distribution. In other words, that argument already assumes the conclusion that Bohr and company wish to reach. See ibid., p. 195, for more on this.

Einstein hoisted by his own petard: See, for example, Kumar 2008 for a "traditional" account of this encounter, in which Einstein is laid low by his own theory, wielded by a victorious Bohr.

"for a totally different purpose": Don Howard 1990, "'Nicht sein kann was nicht sein darf,' or the Prehistory of EPR, 1909–1935: Einstein's Early Worries About the Quantum Mechanics of Composite Systems," in *Sixty-Two Years of Uncertainty: Historical, Philosophical, and Physical Inquiries into the Foundations of Quantum Mechanics*, edited by Arthur I. Miller, 61–111 (Plenum Press). Quote appears on p. 98.

missed the point: Even if Einstein's concern really had been the uncertainty principle, Bohr's invocation of general relativity is alarming rather than ironic. The logical consistency of quantum physics shouldn't rely on the existence of general relativity, as the two theories are not only independent but famously incompatible. There is a resolution of the paradox that Bohr attributed to Einstein that does not involve anything other than quantum physics, but that resolution was not provided by Bohr, and indeed was not provided by anyone at all for decades. See